

**TABLE 1: HOW AI ENHANCES EDGE COMPUTING SERVICES**

Feature	Description	AI-based Approach
Low latency	Minimising the delay in data transmission for real-time applications	Implementing AI algorithms for edge processing, reducing round-trip times
Network slicing	Dividing a physical network into multiple virtual networks for different use cases	AI-driven dynamic network slicing for optimised resource allocation based on demand
Massive IoT connectivity	Connecting a vast number of IoT devices efficiently	AI-powered IoT device management for intelligent connectivity and resource allocation
Enhanced mobile broadband	Providing higher data rates for improved mobile connectivity	AI for optimising spectrum usage, predictive maintenance, and bandwidth allocation
Beamforming	Focusing radio signals in specific directions to enhance signal strength	AI algorithms for dynamic beamforming adjustments based on user locations and network conditions
Network security	Protecting against cyber threats and ensuring data integrity	AI-based anomaly detection, predictive analysis, and real-time threat identification
Resource allocation	Efficiently distributing network resources based on demand	Machine learning for dynamic resource allocation, optimising performance in real time
Energy efficiency	Minimising energy consumption for sustainable network operation	AI-driven energy management, optimising power usage based on traffic patterns and demand
Quality of service (QoS)	Ensuring reliable and high-quality communication services	AI-based monitoring and adaptive adjustments for maintaining optimal QoS levels
Network management	Overseeing and optimising the overall network infrastructure	AI-driven autonomous network management for self-healing, configuration, and optimisation
Predictive maintenance	Anticipating and preventing network equipment failures	AI algorithms analysing historical data for predictive maintenance, reducing downtime
Edge computing integration	Incorporating edge computing for faster data processing at the network edge	AI for efficient task offloading, distributed processing, and edge-to-cloud collaboration
Dynamic network optimisation	Adapting the network in real time to changing conditions and demands	AI-driven algorithms continuously optimising network parameters based on evolving scenarios

constraint. Implementing AI algorithms on edge devices can result in substantial energy consumption, which can have adverse effects on the battery life of mobile devices and increase the operational expenses of IoT devices.

The task of updating and maintaining artificial intelligence models on periphery devices can present significant difficulties. Frequently, cloud-based solutions offer enhanced simplicity in administration for model updates and version control.

The capabilities of peripheral devices can be enhanced through the integration of peripheral AI with 5G technology. The seamless environment created by 5G's increased

network capacity, faster data transfer rates, and decreased latency is ideal for edge AI applications. The convergence of edge AI and 5G technology yields significant effects in situations that demand exceedingly brief response times, including but not limited to connected vehicles, smart cities, and industrial automation.

Decentralised model training is a collaborative learning approach where the training process is distributed across multiple edge devices or nodes instead of relying on a centralised server. In traditional centralised training, all the raw data is sent to a central server, where the model is trained. However, in a

decentralised model, training occurs locally on individual devices, and only model updates are shared.

The rise of edge computing and the advent of 5G networks have made decentralised model training more feasible and efficient. Edge devices, equipped with sufficient computational power, can now participate in training models collectively.

Key components of decentralised model training

Local training. Each edge device performs local model training using its own dataset. This dataset may contain unique patterns and characteristics specific to that device's environment or user behaviour.

Model update sharing. After local training, only the model updates, rather than the raw data, are shared among the devices. This ensures data privacy and security, as sensitive information remains on the local device.

Aggregation and updating. The central server, or a coordinating entity, aggregates the model updates received from various devices. The global model is then updated based on the aggregated information, reflecting the collective knowledge learned from the decentralised network.

Federated learning

Decentralised model training is closely related to federated learning, a subset of this approach. In federated learning, the model is trained across decentralised edge devices, and the global model is iteratively improved without exchanging raw data. This iterative process allows for continuous learning and adaptation to local conditions.

To elaborate further, federated learning extends the notion of decentralised model training by permitting edge devices to educate